

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: HMT500(260313A)_connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: HMT500(260313A)_power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, reset, internal temperature sensor

File: HMT500(260313A)_mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220)

File: HMT500(260313A)_measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: HMT500(260313A)_analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: HMT500(260313A)_rs485.kicad_sch

DOTECH HMT500(260313) – Oil Moisture Transmitter – Schematic v0.8 – Project No. HMT500(260313A)

File names (schematic, PCB, Gerber) = HMT500(260313A).* PCB silkscreen marking = HMT500(260313A)

v0.5: J1 = JST GH 1.25 mm 8P side entry (SM08B–GHS–TB), harness W–2 to M12 8P; J5 chassis spring contact (was M12 shell pin).

v0.4: J3 = JST SH 1.0 mm 4P (SM04B–SRSS–TB side entry, v0.5) for sensor harness W–1 direct from HTX99R (no feedthrough, mech Rev F); Pt1000 2-wire (Kelvin split at J3 pads).

v0.3: MCU STM32G0B1CCT3 (LQFP48, –40..125 C, in-house part); TPS26600PWP official pinout; single axial PCB 57x23 (mech Rev C).

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS–485 Modbus RTU + 2x analog (4–20 mA / 0–20 mA / 0–10 V / 0–5 V selectable). Supply 12–28 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

v0.6: parts cleanup (–16: pull-ups R11–15, LED D3/R16, RS–485 R60/61 D60/61, FB2 C19 C11 C2 C6); PCB temperature = ADS1220 + MCU internal sensors.

v0.7: DAC8760 datasheet pinout (HTSSOP–24 PWP); +VSENSE via OPA197 follower (U12/U13); external ISET–R 15k 0.1%; DACs on SPI2 daisy chain, common DAC_LATCH.

v0.8: datasheet review (docs/hw/schematic–review–260313A.md): real pinouts LMR36006/PCAP04/TPS26611; TPS2660 RTN isolated from GND; OVP 29 V (supply 12–28 V);

DAC SPI = gated SCLK (daisy chain removed in DAC8760 rev D); DAC AVDD 10R; ALARM wired–OR + 10k; TVS3301 outputs; PCAP04 on SPI3; buck L/C per datasheet.

TBD footprints: J5 chassis contact, CMC, GDT, buck inductor L2, LMR36006 (RXN0012A 2x3 mm, not in KiCad lib).

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울임 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPI_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

글자 편집: 생성기가 부품 몸체 핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

Pinouts checked vs manufacturer datasheets 2026–09–30 (STM32: KiCad lib)

Design notes: docs/hw/circuit–design.md

Project No. HMT500(260313A) – HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /

File: HMT500(260313A).kicad_sch

Title: HMT500(260313A) Top

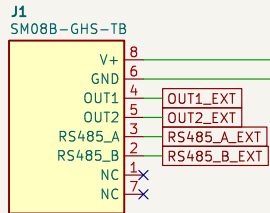
Size: A3Date: 2026–09–30Rev: 0.8

KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4Id: 1/7

FIELD CONNECTOR

현장 커넥터

J1 JST GH 8P -> W-2 -> M12 8P
EE364 pinout, pin n = pin n

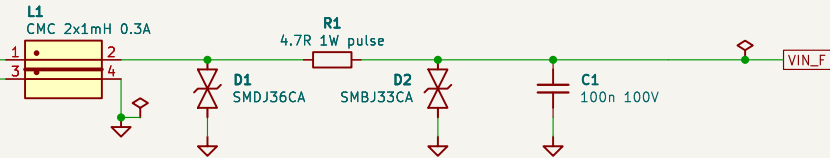


Pins 1, 7: not connected
GH mounting pads (MP): PCB only

INPUT SURGE / REVERSE-POLARITY PROTECTION

입력 서지·역극성 보호

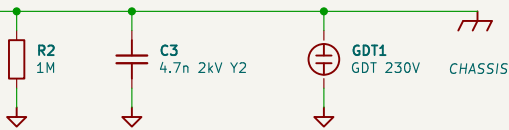
TVS bidirectional: -30 V miswiring must not conduct



SMDJ 3 kW (1st) -> R1 -> SMBJ (2nd) -> eFuse on sheet Power

CIRCUIT GND <--> CHASSIS (floating)

회로 GND-외함 분리



PCB-housing creepage >= 2 mm

GDT conducts only on line-to-ground surge

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
J1	SM08B-GHS-TB	현장 커넥터 연결	JST GH 1.25 mm 8P 옆 삽입, 하네스 W-2 → M12 8P, 핀 n = M12 핀 n
L1	CMC 2x1mH 0.3A	전원선 공통모드 노이즈 차단	2 × 1 mH, 0.3 A — 전도 방출·전도 내성 대책
D1	SMDJ36CA	입력 서지 1단 흡수	TVS 양방향 3000 W, 36 V — -30 V 오결선에 도통 안 함
R1	4.7R 1W pulse	1단·2단 서지 분담	MELF 1 W 펄스 내량 저항
D2	SMBJ33CA	입력 서지 2단 클램프	TVS 양방향 600 W, 33 V — eFuse 정격 이하로 제한
GDT1	GDT 230V	회로 GND-외함 서지 방전	230 V 방전관, 선-대지 서지 때만 도통 (평소 절연)
C3, R2	4.7n Y2 / 1M	GND-외함 고주파 결합	Y2 안전 콘덴서 + 1 MΩ 정전기 방전 경로
J5	Chassis contact	PCB-하우징 겹지 접점	SMD 스프링 접점, PCB 가장자리 → 하우징 Ø27 내면

* 표시 값은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

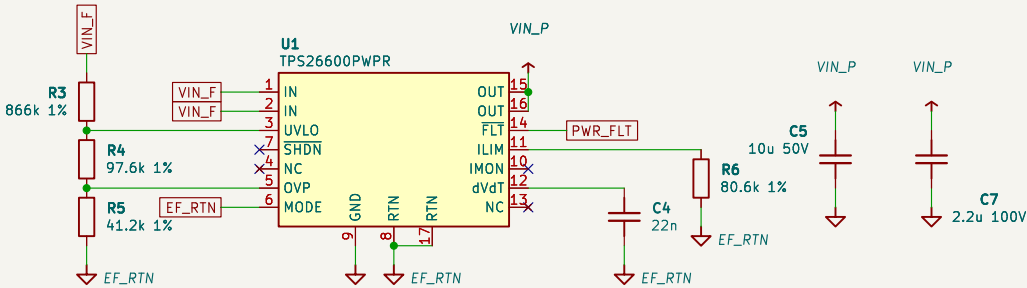
Sheet: /1. Connector & input protection/
File: HMT500(260313A)_connector.kicad_sch

Title: HMT500(260313A) Connector & input protection

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

eFuse TPS2660 reverse -60 V / OVP 29 V / UVLO 8.6 V

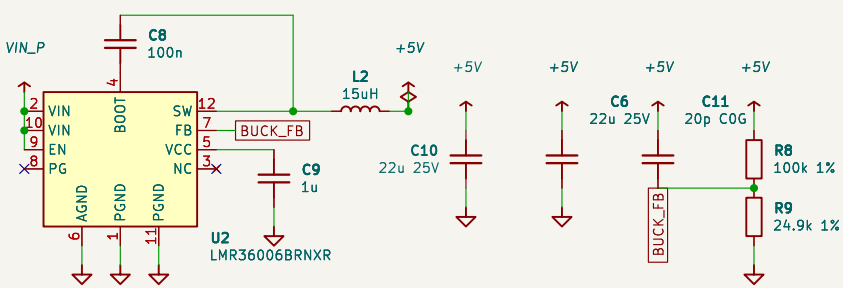
전자 퓨즈



$UVLO = 1.19 \text{ V} \times (R3+R4+R5)/(R4+R5) = 8.6 \text{ V}$ $OVP = 1.19 \text{ V} \times (R3+R4+R5)/R5 = 29.0 \text{ V}$ (28.5-29.9)
 $I_{OL} = 12 / R6[\text{kohm}] = 149 \text{ mA}$. Supply range 12-28 V: TPS26611 +Vs max 30 V (U9/U10 powered from VIN_P).
EF_RTIN = TPS2660 RTN reference: R5, R6, C4, MODE, PowerPAD. NEVER connect to GND (datasheet 9.3.5.5).

BUCK 5 V LMR36006 (4.2-60 V in, 0.6 A)

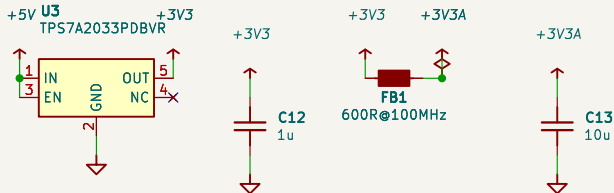
5 V 강압 전원



$V_{out} = 1.0 \text{ V} \times (1 + R8/R9) = 5.0 \text{ V}$. 1 MHz: L 15 uH, COUT 2 x 22 uF, CFF 20 pF (datasheet Table 1)

LDO 3.3 V TPS7A2033 + analog rail +3V3A

3.3 V 레귤레이터



주요 부품 — 역할·특징·사양

부품번호	부품	역할	특징·사양
U1	TPS26600PWPR	전자 퓨즈 (입력 보호)	역극성 -60 V 차단, 과전압 29 V·저전압 8.6 V (R3-R5), 전류 제한 149 mA (R6), 돌입 제한 (C4), 고장 출력 FLT
EF_RTIN	(기준 접지)	TPS2660 내부 기준	R5·R6·C4·MODE·방열 패드는 RTN에 연결. GND와 연결 금지 — 역극성 보호 무효·손상
U2	LMR36006BRNXR	5 V 강압 전원 (벅)	입력 4.2-60 V, 0.6 A 동기식, 1 MHz, 출력 = $1 \text{ V} \times (1 + R8/R9)$
L2, C6, C10, C11	15uH / 22u×2 / 20p	벅 출력 필터 보상	데이터시트 권장값 (5 V, 1 MHz). L2 4×4 mm 차폐형, 포화 0.8 A 이상
U3	TPS7A2033PDBVR	3.3 V 저잡음 레귤레이터	300 mA, 저잡음·높은 PSRR → 벅 리플 제거, SOT-23-5
FB1	600R@100MHz	아날로그 전원 +3V3A 분리	페라이트 비드 + C13 10 uF → PCAP04-ADS1220 전원

• 표시 값은 데이터시트로 확인

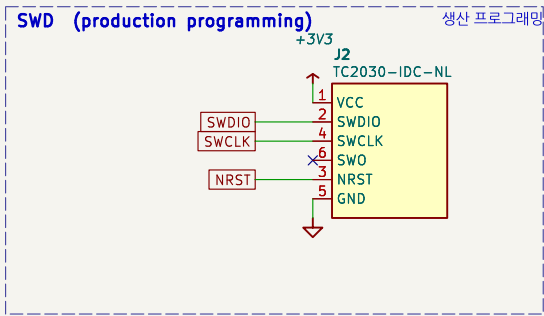
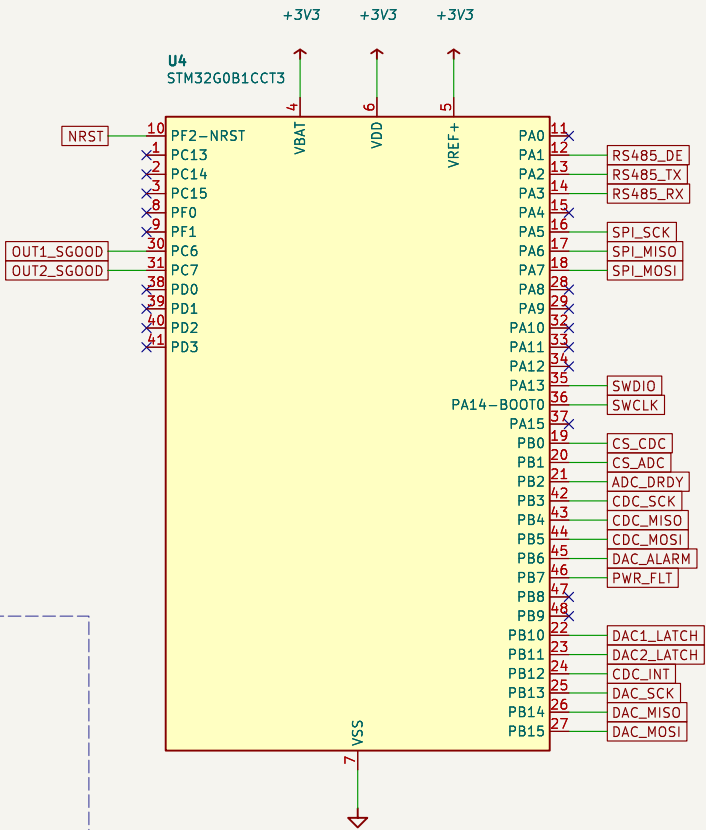
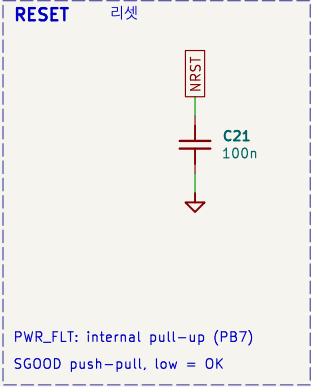
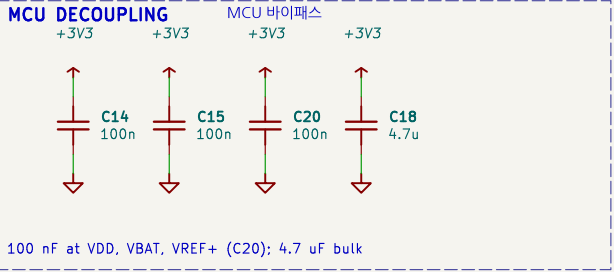
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) — HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /2. Power/
File: HMT500(260313A)_power.kicad_sch

Title: HMT500(260313A) Power

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/7



MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

-> boot from flash (RS-485 bootloader 0x08000000).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

주요 부품 — 역할·특징·사양

부품번호	부품	역할	특징·사양
U4	STM32G0B1CCT3	제어·통신·보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-125 °C, LQFP48, 내부 온도센서 (PCB 온도). SPI1 = ADS1220, SPI2 = DAC (모드 3, SCLK 게이트), SPI3 = PCAP04
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유

* 표시 값은 데이터시트로 확인

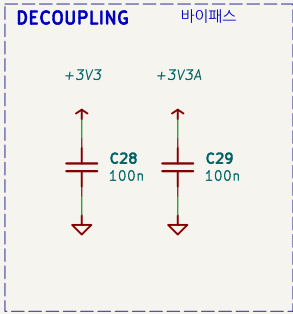
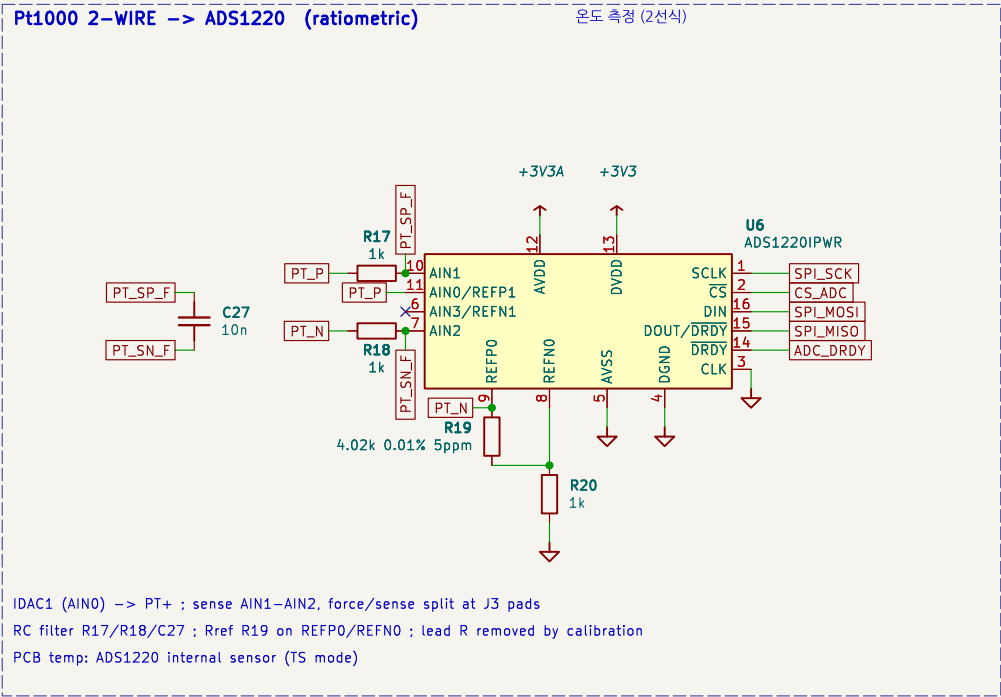
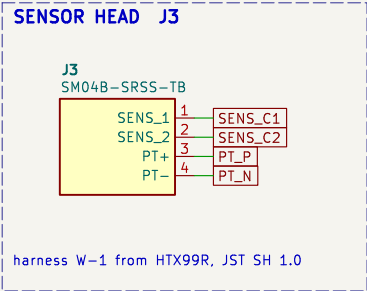
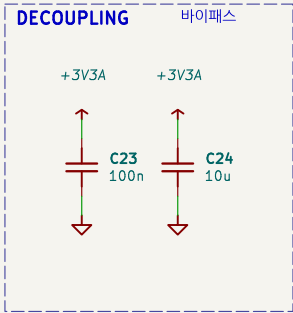
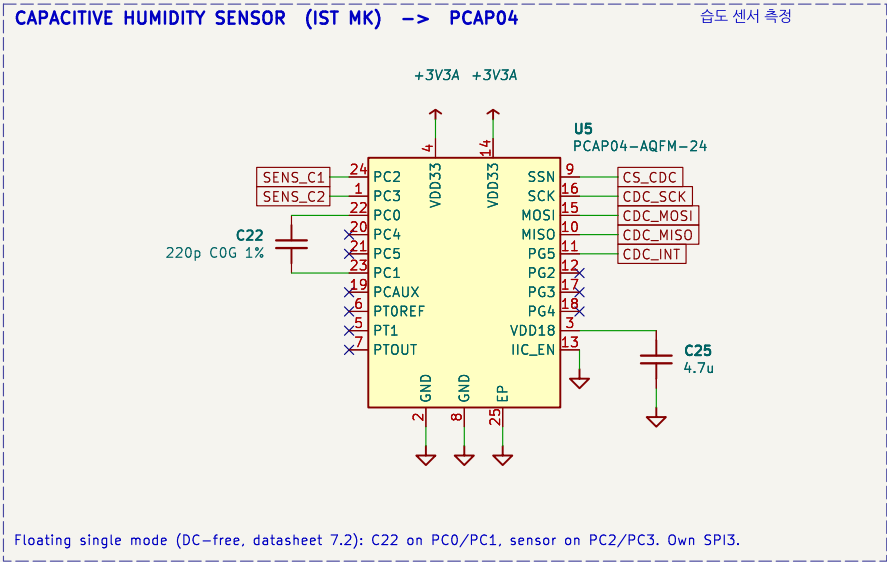
Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter

DOTECH Co., Ltd.

Sheet: /3. MCU/
File: HMT500(260313A)_mcu.kicad_sch

Title: HMT500(260313A) MCU

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/7



주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
J3	SM04B-SRSS-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1-2 = MK33, 3-4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정(방전 시간), DC 없는 구동, 내부 DSP, 전용 SPI3 (모드 1)
C25, C24	4.7u / 10u	PCAP04 전원 버퍼	VDD18 ≥ 4.7 uF, VDD33 ≥ 10 uF (데이터시트 표 2)
C22	220p COG 1%	PCAP04 기준 용량	MK33-W mini (200 pF)와 비슷한 값, COG (온도 변화 거의 없음)
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도), 설정 한계: IDAC ≤ 250 uA, PGA ≤ 2 (SPI1 모드 1)
R19	4.02k 0.01% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n	입력 RC 필터	차동-공통모드 노이즈 제거
* 표시 값은 데이터시트로 확인			

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
NOTECH Co., Ltd.

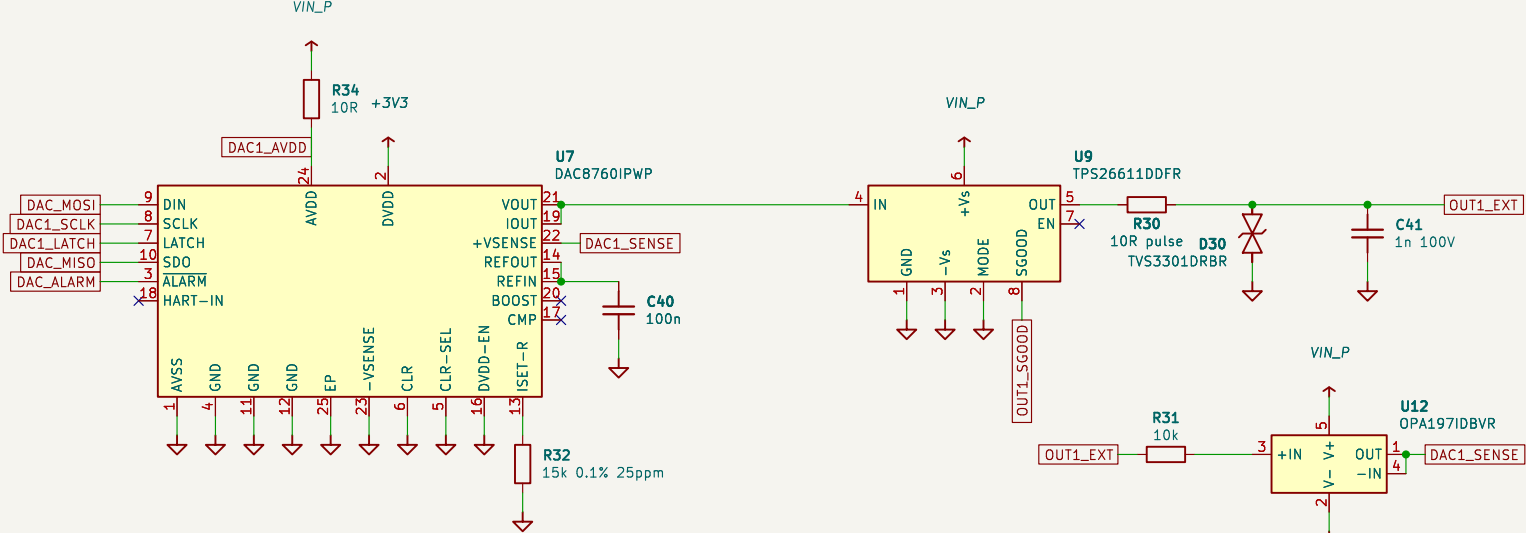
Sheet: /4. Measurement/
File: HMT500(260313A)_measurement.kicad_sch

Title: HMT500(260313A) Measurement

Size: A3	Date: 2026-09-30	Rev: 0.8
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/7

ANALOG OUTPUT CH1 - V/I selectable, miswiring protected -> connector pin 4

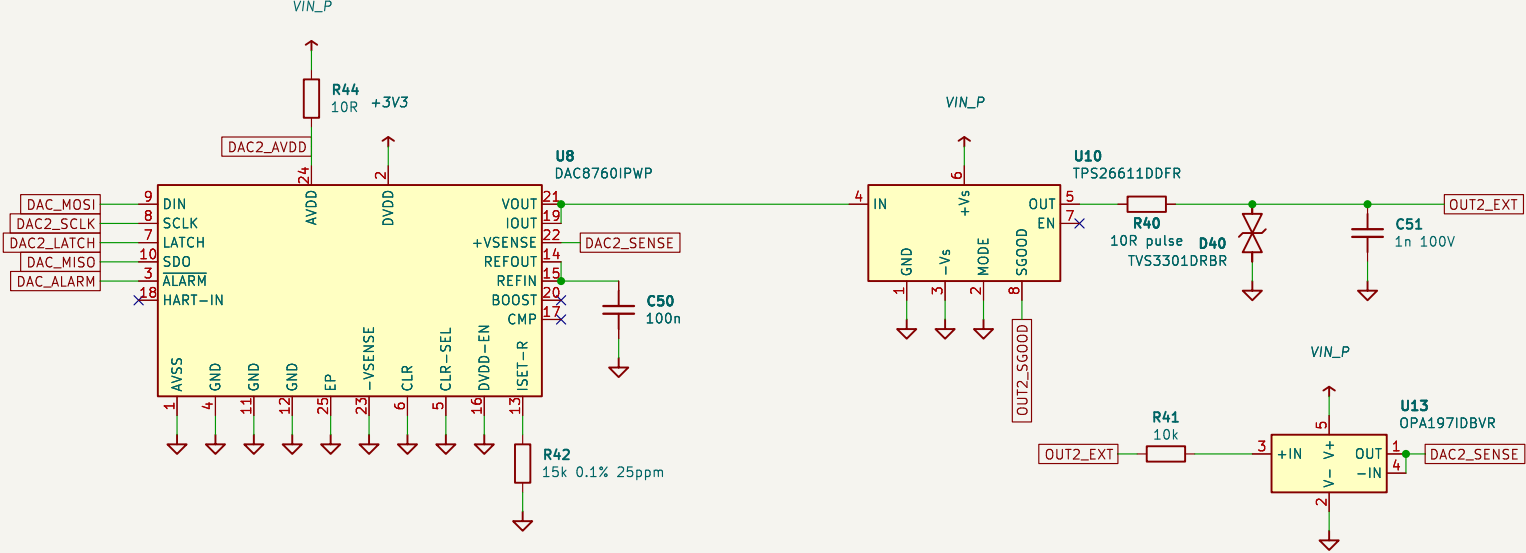
아날로그 출력 1



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U12 OPA197 follower of terminal OUT1_EXT (after R30) via R31:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R32 15k 0.1%. SGOOD low = OK.

ANALOG OUTPUT CH2 - V/I selectable, miswiring protected -> connector pin 5

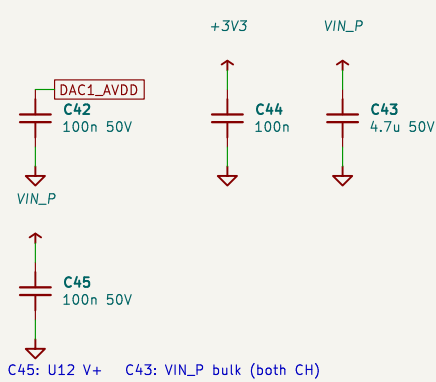
아날로그 출력 2



VOUT + IOUT one terminal (datasheet 9.1.1.3). +VSENSE = U13 OPA197 follower of terminal OUT2_EXT (after R40) via R41:
no +VSENSE 60k leakage in current mode, series-R drop corrected in voltage mode. ISET-R = R42 15k 0.1%. SGOOD low = OK.

CH1 DECOUPLING

채널 1 바이패스



DAC SELECTION (decision #10)

DAC 선정

U7/U8 fitted: DAC8760 (16-bit, I + V output).

Drop-in alternates (same pinout, no layout change):

DAC7760 12-bit, I + V (cost-down, V/I kept)

DAC8750 16-bit, I only 4-20/0-20/0-24 mA (EE364-equiv.)

DAC7750 12-bit, I only (~5 uA/step vs EE364 2 uA)

Current-only (x750): VOUT/+VSENSE/-VSENSE absent -

check pin handling before fitting (VERIFY).

DAC SPI - SPI2, SCLK gated per DAC

DAC 전용 SPI

DAC8760 rev D: daisy chain removed; multiple devices

need gated SCLK (datasheet 8.5.1.5, Fig. 8-9).

SPI2 mode 3 (CPOL 1, CPHA 1): SCK idles high, so the

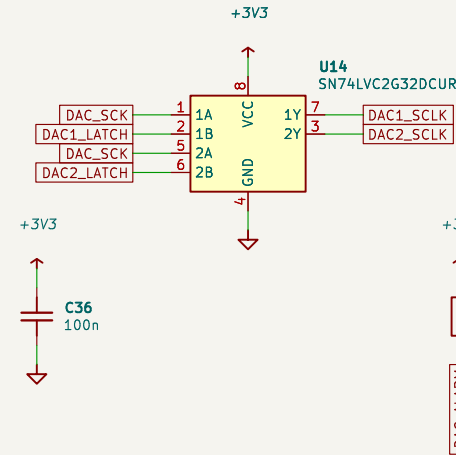
OR gate gives no extra edge at LATCH low/high.

DACn_SCLK = DAC_SCK OR DACn_LATCH (PB10 / PB11).

DIN and SDO shared (SDO 3-state after LATCH).

ALARM: open drain, wired-OR, R33 10k pull-up:

which channel -> read status register.



주요 부품 - 역할·특징·사양

부품번호	부품	역할	특징·사양
U7, U8	DAC8760IPWP	아날로그 출력 1·2	16비트, 4-20 / 0-20 / 0-24 mA · 0-5 / 0-10 V 선택, 내부 기준, 고정 알람 - 대체 DAC7760/8750/7750
U9, U10	TPS26611DDFR	출력 오걸선 보호	±50 V, 단자 > +Vs 또는 < -0.2 V면 5 us 차단·자동 복귀, 전류 제한 32 mA, RON 7.5 Ω, SGOOD Low = 정상
R30, R40	10R pulse (2512)	서지 전류 제한	출력 보호기와 TVS 사이 직렬
D30, D40	TVS3301DRBR	출력 서지 클램프	TI 평탄 클램프 33 V 양방향, 40 V @ 27 A - TPS26611 ±55 V 보호 (데이터시트 권장)
C41, C51	1n 100V	출력 고주파 필터	케이블로 들어오는 RF 억제
U12, U13	OPA197IDBVR	+VSENSE 버퍼 (이득 1)	36 V, 입출력 레일투레일 - 단자 전압을 +VSENSE로, 전류 모드 누설 0
R31, R41	10k	버퍼 입력 보호	R30-R40 뒤 단자 전압을 OPA197 + 입력으로 (서지 전류 제한)
R32, R42	15k 0.1% 25ppm	외부 ISET-R (전류 기준)	내부 저항보다 온도 드리프트 작음 (REXT 비트 설정)
R34, R44	10R	DAC AVDD 직렬	전원 상승 속도 제한 (데이터시트 10장)
U14	SN74LVC2G32DCUR	DAC별 SCLK 게이트	OR 2회로: DACn_SCLK = SCK OR LATCHn (SPI2 모드 3)
R33	10k	ALARM 플업	오픈 드레인 ALARM 두 개를 한 선으로 (데이터시트: 외부 10 kΩ)

* 표시 같은 데이터시트로 확인

Pinouts checked vs manufacturer datasheets 2026-09-30 (STM32: KiCad lib)
Design notes: docs/hw/circuit-design.md
Project No. HMT500(260313A) - HMT500(260313) oil moisture transmitter
DOTECH Co., Ltd.

Sheet: /5. Analog outputs/
File: HMT500(260313A)_analog_out.kicad_sch

Title: HMT500(260313A) Analog outputs

Size: A3 Date: 2026-09-30

Rev: 0.8

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Id: 6/7

